WORKED SOLUTION

STEP I 1990, Question 10

Mechanics • Power and Resistance

STEP I 1990, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q10

The key idea

"Steady speed" means no acceleration, so the runner's driving force exactly balances the resistances, and power is driving force times speed. The only subtlety is that air resistance depends on the speed *relative to the air*, which is different on the two legs. For the second half of the question, think about total energy rather than power: the energy spent on each leg is resistance times distance, and that turns the fixed energy budget into a linear condition on the two speeds.

Solution

Steady speeds and the time for the race

Model the runner as a particle moving along a straight line. Let v (in $\mathrm{m\,s^{-1}}$) be the runner's ground speed on a leg and D (in N) the forward driving force the runner produces. The resistances are $50\,\mathrm{N}$ and $2w\,\mathrm{N}$, both opposing motion relative to the ground and the air respectively.



Resolving along the track, the acceleration is zero at steady pace, so

$$D = 50 + 2w.$$

The rate at which the runner does work is the driving force times the speed of its point of application, so $300 = Dv$.

Outward leg. The wind blows at $3\,\mathrm{m\,s^{-1}}$ in the runner's direction of motion. If $v > 3$ the runner moves forward relative to the air at $w = v - 3$, and air resistance acts backwards. (We check $v > 3$ below.) Then $D = 50 + 2(v-3) = 44 + 2v$, and

$$v_1(44 + 2v_1) = 300 \iff v_1^2 + 22v_1 - 150 = 0 \iff v_1 = -11 + \sqrt{271} \approx 5.462,$$

taking the positive root. This is indeed greater than 3, so the assumption holds. (Alternatively: at $v = 3$ there is no air resistance and the power needed is only $3 \times 50 = 150 < 300$ W, so the runner must go faster than the wind.)

Return leg. Now the runner runs into the wind, so $w = v + 3$ and $D = 50 + 2(v + 3) = 56 + 2v$:

$$v_2(56 + 2v_2) = 300 \iff v_2^2 + 28v_2 - 150 = 0 \iff v_2 = -14 + \sqrt{346} \approx 4.601.$$

Time. Ignoring the acceleration at the start and turn,

$$T_0 = \frac{5000}{v_1} + \frac{5000}{v_2} \approx 915.4 + 1086.7 = 2002.1,$$

so the race takes 2002 s to the nearest second (about 33 minutes 22 seconds).

Alternative tactics

Suppose instead the runner works at x_1 W outward and x_2 W on the return, with steady speeds v_1 and v_2. The force balance is unchanged, so as long as $v_1 > 3$,

$$x_1 = v_1(44 + 2v_1), \qquad x_2 = v_2(56 + 2v_2).$$

The time on the outward leg is $5000/v_1$, so the energy used there is

$$x_1 \cdot \frac{5000}{v_1} = 5000(44 + 2v_1),$$

which is just the resistance times the distance, as it should be. Similarly the return leg uses $5000(56 + 2v_2)$. The total energy is therefore

$$E = 5000(100 + 2(v_1 + v_2)).$$

Ask yourself Where does the fixed total energy come in?

The runner must spend the same total energy as in the original race, so E is a fixed number. Since E depends on the speeds only through $v_1 + v_2$, that sum is forced:

$$v_1 + v_2 = \frac{E}{10\,000} - 50,$$

whatever the choice of x_1 and x_2. This proves the result. Its value is the one from the original race,

$$S = v_1 + v_2 = \sqrt{271} + \sqrt{346} - 25 \approx 10.063.$$

The best tactics. The time for the race is now

$$T = 5000\left(\frac{1}{v_1} + \frac{1}{v_2}\right) = \frac{5000\,(v_1 + v_2)}{v_1 v_2} = \frac{5000\,S}{v_1 v_2}.$$

With S fixed, this is smallest when $v_1 v_2$ is largest. By the AM–GM inequality, $v_1 v_2 \leq (\frac{S}{2})^2$, with equality exactly when $v_1 = v_2 = \frac{S}{2}$. So the best plan is to run at the same speed both ways, $\frac{S}{2} \approx 5.032\,\mathrm{m\,s^{-1}}$. That is still faster than the wind, so the outward formula for x_1 remains valid. The minimum time is

$$T_{\min} = \frac{5000\,S}{S^2/4} = \frac{20\,000}{S} \approx 1987.4\ \mathrm{s}.$$

The runner saves $T_0 - T_{\min} \approx 2002.1 - 1987.4 = 14.7\,\mathrm{s}$, which is approximately 15 seconds, as required. The powers are about $x_1 \approx 272\,\mathrm{W}$ outward and $x_2 \approx 332\,\mathrm{W}$ back: the runner eases off with the wind behind and works harder into it.

Checks

- Energy consistency: $300 \times 2002.105 \approx 600\,632\,\mathrm{J}$, and $5000(100 + 2 \times 10.0632) \approx 600\,632\,\mathrm{J}$ also.
- With no wind, both legs would give $v^2 + 25v - 150 = 0$, so $v = 5$, and the two methods would coincide. The wind is what makes the saving possible.
- We checked numerically that $v_1 + v_2 = 10.0632$ for 1000 random choices of x_1 (with x_2 fixed by the energy budget), that no choice beats 1987.4 s, and that the saving is 14.66 s.

Common slips

- Using the ground speed in the air-resistance term instead of the speed relative to the air, or getting the sign of the wind the wrong way round on one leg.
- Thinking that equal *power* both ways is fair: it is equal *speed* both ways that minimises the time.
- Trying to prove $v_1 + v_2$ constant from the power equations alone. You need the fixed total energy.

Beyond STEP

The last part is a small constrained optimisation: minimise $1/v_1 + 1/v_2$ subject to a linear constraint. AM–GM settles it here, but the general tool, met in first-year courses, is the method of Lagrange multipliers, and the same ideas govern optimal pacing in cycling and rowing. For a careful treatment of work, power and drag forces, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

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